Exp no:11 Determination and plotting of Magnetic Flux density in Ferromagnetic Materials with different (i) length, (ii) width and (iii) magnetic field intensity.

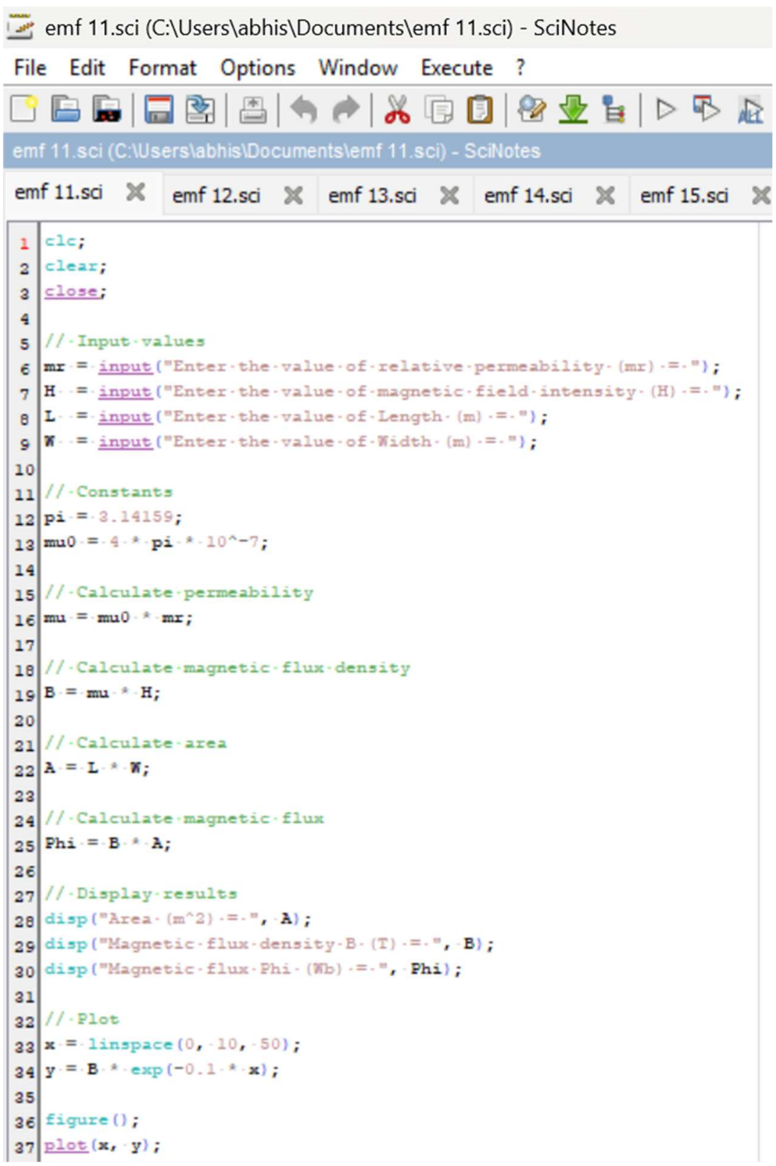
Aim: To write program for Magnetic flux density in ferromagnetic materials using SCILAB.

Software required:

•SCILAB version 6.0.1

Theory:

The magnetic flux density (B) in ferromagnetic materials is a measure of the magnetic field's strengthwithin the material, which takes into account both the external magnetic field (H) and the material'sinherent magnetic properties (i.e., its magnetization). The relationship between these quantities in a ferromagnetic material is more complex than in non-magnetic or paramagnetic materials because




```
clc;
clear;
close;

// Input values
mr = input("Enter the value of relative permeability (mr) = ");
H  = input("Enter the value of magnetic field intensity (H) = ");
L  = input("Enter the value of Length (m) = ");
W  = input("Enter the value of Width (m) = ");

// Constants
pi = 3.14159;
mu0 = 4 * pi * 10^-7;

// Calculate permeability
mu = mu0 * mr;

// Calculate magnetic flux density
B = mu * H;

// Calculate area
A = L * W;

// Calculate magnetic flux
Phi = B * A;

// Display results
disp("Area (m^2) = ", A);
disp("Magnetic flux density B (T) = ", B);
disp("Magnetic flux Phi (Wb) = ", Phi);

// Plot
x = linspace(0, 10, 50);
y = B * exp(-0.1 * x);

figure();
plot(x, y);
```

Scilab 2026.1.0 Console

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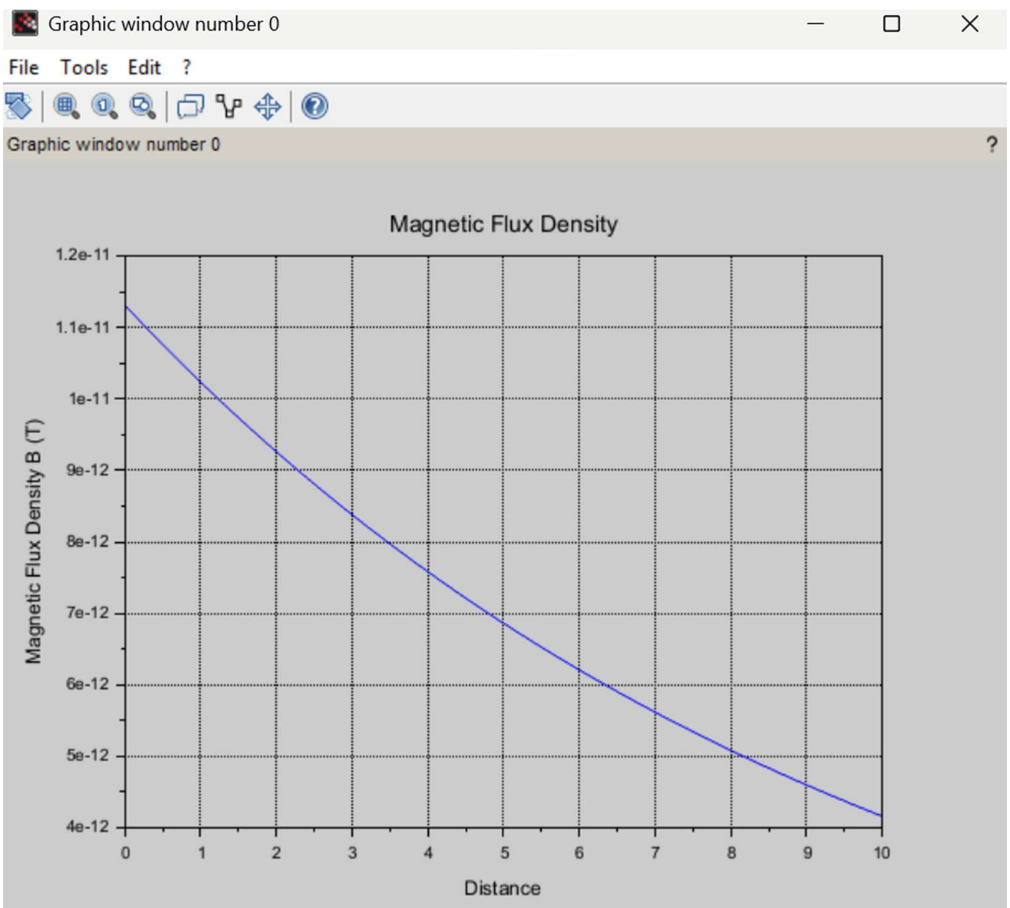
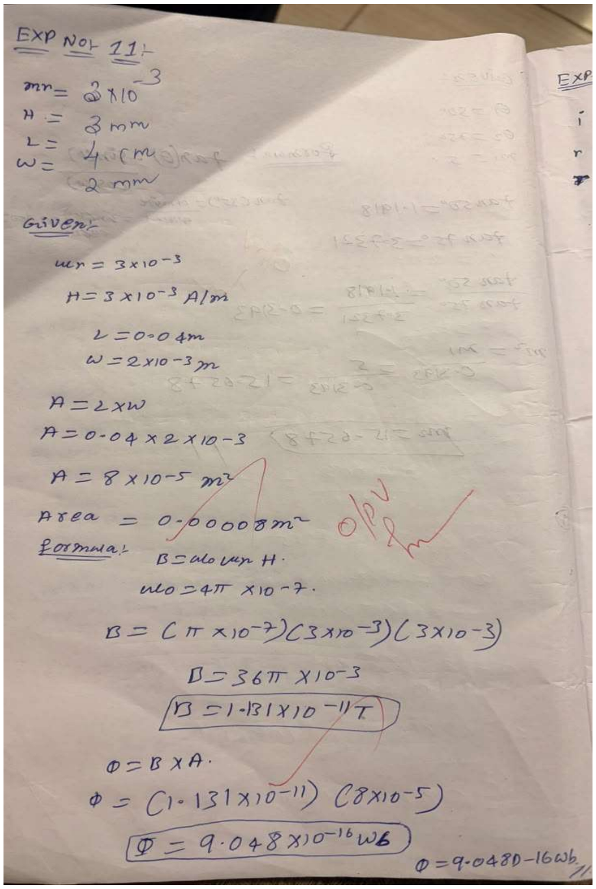
Scilab 2026.1.0 Console

```
Enter the value of relative permeability (mr) = 3*10^-3

Enter the value of magnetic field intensity (H) = 3*10^-3

Enter the value of Length (m) = 0.04

Enter the value of Width (m) = 2*10^-3

  "Area (m^2) = "
   0.00008
  "Magnetic flux density B (T) = "
   1.131D-11
  "Magnetic flux Phi (Wb) = "
   9.048D-16
```





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$mr = 3 \times 10^{-3}$

$H = 3\,mm$

$L = 4\,cm$

$W = 2\,mm$

Given:

$\mu r = 3 \times 10^{-3}$

$H = 3 \times 10^{-3}\ A/m$

$L = 0.04\,m$

$W = 2 \times 10^{-3}\,m$

$A = L \times W$

$A = 0.04 \times 2 \times 10^{-3}$

$A = 8 \times 10^{-5}\ m^2$

Area $= 0.00008\,m^2$

Formula: $B = \mu_0 \mu_r H.$

$\mu_0 = 4\pi \times 10^{-7}.$

$B = (\pi \times 10^{-7})(3 \times 10^{-3})(3 \times 10^{-3})$

$B = 36\pi \times 10^{-3}$

$\boxed{B = 1.131 \times 10^{-11}\,T}$

$\Phi = B \times A.$

$\Phi = (1.131 \times 10^{-11})(8 \times 10^{-5})$

$\boxed{\Phi = 9.048 \times 10^{-16}\,Wb}$

$\Phi = 9.0480 - 16\,Wb$

Case 1: Examine the effect of length of a ferromagnetic material on the magnetic flux density (B).

Analytical :

Calculate the magnetic flux density for the following conditions

 The material is ferromagnetic with relative permeability (μr) of 500.

 The magnetic field intensity (H) is held constant at 100 A/m.

 The length of the material varies from 0.1 m to 2 m.

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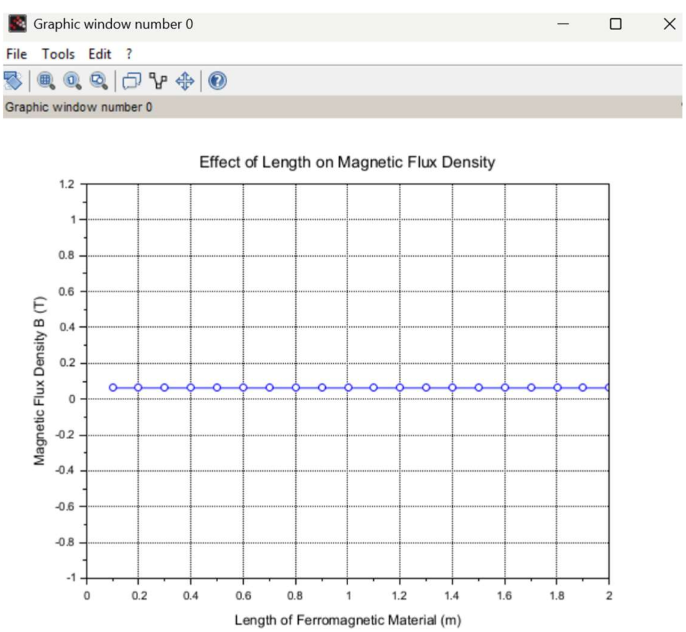
```
clc;
clear;
clf;

// Constants
mu_r = 500;              // Relative permeability
H = 100;                 // Magnetic field intensity (A/m)
lengths = 0.1:0.1:2;     // Lengths from 0.1 m to 2 m
mu_0 = 4 * %pi * 10^-7;  // Permeability of free space

// Initialise an array to store magnetic flux density values
B_values = zeros(1, length(lengths));

// Calculate magnetic flux density for each length
for i = 1:length(lengths)
    B_values(i) = mu_r * mu_0 * H;
end

// Display results
disp("Length (m)       Magnetic Flux Density (T)");

for i = 1:length(lengths)
    disp(string(lengths(i)) + "             " + string(B_values(i)));
end

// Plotting the results
plot(lengths, B_values, "-o");

xlabel("Length of Ferromagnetic Material (m)");
ylabel("Magnetic Flux Density B (T)");
title("Effect of Length on Magnetic Flux Density");

xgrid();
```



Case 2: To investigate how the magnetic flux density (B) in a ferromagnetic material changes with

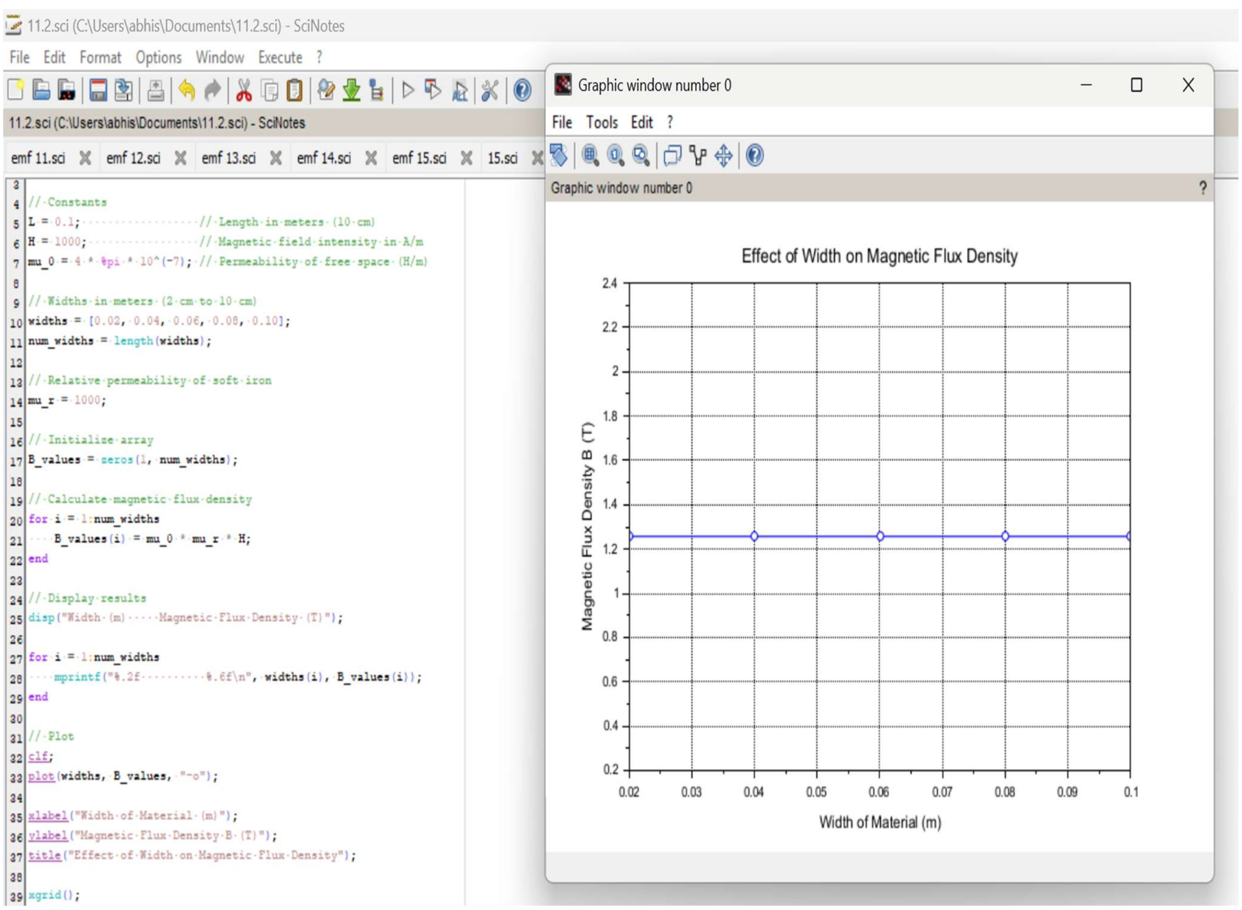
different widths at a constant length and magnetic field intensity.

Analytical:

Calculate magnetic flux density (B) in a ferromagnetic material changes with different widths at a constant length

and magnetic field intensity for the following condtions

- Material: Soft iron (same as in Case Study 1)
- Length (L): Constant at 10 cm
- Magnetic Field Intensity (H): Constant at 1000 A/m
- Width (W): Varies from 2 cm to 10 cm (i.e., 2 cm, 4 cm, 6 cm, 8 cm, 10 cm)



Case 3: To examine how the magnetic flux density (B) in a ferromagnetic material change with different

values of magnetic field intensity at constant length and width.

Analytical:

Calculate the magnetic flux density (B) in a ferromagnetic material change with different values of

magnetic field intensity at constant length and width.

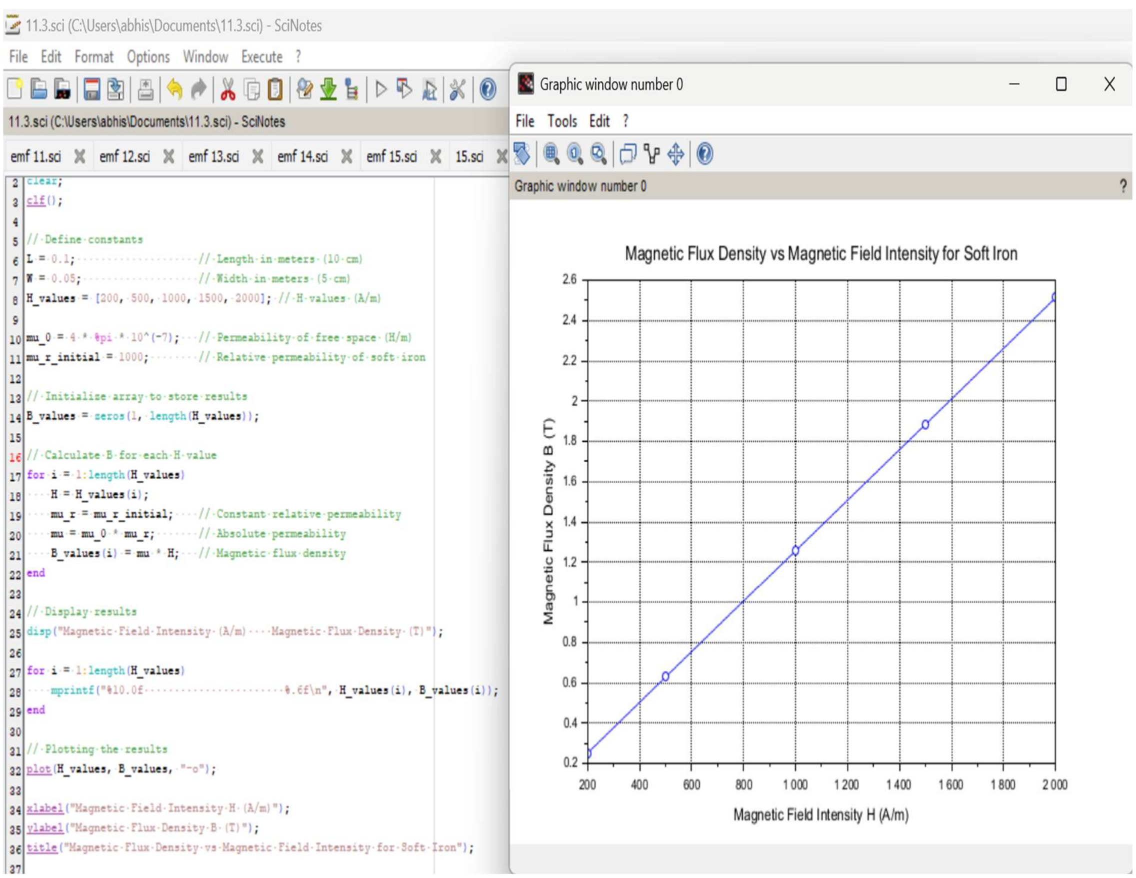
 Material: Soft iron (same as in the previous studies)

 Length (L): Constant at 10 cm

 Width (W): Constant at 5 cm

 Magnetic Field Intensity (H): Varies from 200 A/m to 2000 A/m (i.e., 200 A/m, 500 A/m, 1000

A/m, 1500 A/m, 2000 A/m)



Result :Thus the program was verified for program for Magnetic flux density in ferromagnetic materials using SCILAB was successfully.